

**BOLUNGAVIK, Iceland**WMO: **040050**Lat: **66.161N**Lon: **23.254W**Elev: **89**StdP: **14.65**Time Zone: **0.00 (E00)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

| Coldest Month | Heating DB |      | Humidification DP/MCDB and HR |     |      |     |     |      | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|------|-------------------------------|-----|------|-----|-----|------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |      | 99.6%                         |     |      | 99% |     |      | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%  | DP                            | HR  | MCDB | DP  | HR  | MCDB | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)  | (d)                           | (e) | (f)  | (g) | (h) | (i)  | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 2             | 15.1       | 17.8 | 7.0                           | 7.9 | 17.0 | 9.8 | 9.1 | 19.4 | 45.3                  | 29.8 | 40.3 | 31.2 | 11.5                  | 240  | 0.801 |

**Annual Cooling, Dehumidification, and Enthalpy Design Conditions**

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 8.6                    | 60.8            | 53.3 | 58.7 | 52.2 | 56.8 | 51.1 | 54.7                | 59.1 | 53.1 | 57.3 | 52.0 | 56.0 | 8.1                  | 20   |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 52.0                            | 57.8 | 56.6 | 50.6 | 54.8 | 55.3 | 49.4 | 52.3 | 54.2 | 23.0          | 59.3 | 22.2 | 57.6 | 21.5 | 56.1 | 63.0              |

**Extreme Annual Design Conditions**

| Extreme Annual WS |             |             | Extreme Annual Temperature |             |                    |            | n-Year Return Period Values of Extreme Temperature |             |            |             |            |             |            |             |  |  |
|-------------------|-------------|-------------|----------------------------|-------------|--------------------|------------|----------------------------------------------------|-------------|------------|-------------|------------|-------------|------------|-------------|--|--|
|                   |             |             | Mean                       |             | Standard Deviation |            | n=5 years                                          |             | n=10 years |             | n=20 years |             | n=50 years |             |  |  |
| 1%                | 2.5%        | 5%          | Min                        | Max         | Min                | Max        | Min                                                | Max         | Min        | Max         | Min        | Max         | Min        | Max         |  |  |
| (a)               | (b)         | (c)         | (e)                        | (f)         | (g)                | (h)        | (i)                                                | (j)         | (k)        | (l)         | (m)        | (n)         | (o)        | (p)         |  |  |
| (4) <b>35.3</b>   | <b>30.9</b> | <b>26.9</b> | <b>11.9</b>                | <b>65.4</b> | <b>3.6</b>         | <b>2.8</b> | <b>9.3</b>                                         | <b>67.4</b> | <b>7.2</b> | <b>69.0</b> | <b>5.1</b> | <b>70.6</b> | <b>2.5</b> | <b>72.6</b> |  |  |
| (5)               |             |             | <b>10.2</b>                | <b>58.0</b> | <b>3.3</b>         | <b>2.1</b> | <b>7.9</b>                                         | <b>59.5</b> | <b>5.9</b> | <b>60.8</b> | <b>4.1</b> | <b>61.9</b> | <b>1.7</b> | <b>63.5</b> |  |  |

**Monthly Climatic Design Conditions**

|                                                                               |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|-------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                               |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                           | DBAvg       | 38.9   | 32.0  | 30.9  | 31.2  | 34.9  | 40.3  | 46.8  | 50.2  | 49.5  | 45.4  | 38.5  | 34.3  | 32.2 |
|                                                                               | DBStd       | 9.07   | 6.26  | 6.68  | 6.77  | 5.83  | 5.08  | 4.59  | 4.01  | 3.98  | 5.69  | 5.93  | 5.96  | 6.49 |
|                                                                               | HDD50       | 4194   | 558   | 534   | 584   | 453   | 303   | 115   | 47    | 57    | 161   | 359   | 471   | 552  |
|                                                                               | HDD65       | 9529   | 1023  | 954   | 1049  | 903   | 766   | 545   | 460   | 480   | 589   | 822   | 921   | 1017 |
|                                                                               | CDD50       | 140    | 0     | 0     | 0     | 0     | 2     | 19    | 52    | 43    | 22    | 2     | 0     | 0    |
|                                                                               | CDD65       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH74       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
|                                                                               | CDH80       | 0      | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0     | 0    |
| Wind                                                                          | WSAvg       | 10.4   | 12.2  | 12.3  | 12.1  | 10.6  | 9.2   | 8.5   | 7.3   | 7.7   | 9.2   | 11.8  | 11.8  | 12.5 |
| Precipitation                                                                 | PrecAvg     | 43.0   | 5.0   | 4.7   | 4.1   | 2.6   | 1.9   | 1.3   | 1.8   | 2.4   | 4.1   | 5.0   | 5.0   | 5.3  |
|                                                                               | PrecMax     | 64.8   | 8.9   | 13.0  | 7.0   | 5.2   | 4.4   | 3.7   | 4.6   | 4.3   | 6.5   | 8.3   | 8.2   | 9.6  |
|                                                                               | PrecMin     | 32.0   | 2.2   | 1.7   | 1.4   | 1.0   | 0.5   | 0.0   | 0.6   | 0.7   | 1.3   | 1.7   | 2.2   | 1.4  |
|                                                                               | PrecStd     | 7.4    | 1.7   | 2.3   | 1.4   | 1.1   | 1.0   | 0.8   | 0.9   | 1.1   | 1.5   | 1.9   | 1.7   | 1.8  |
| Monthly Design<br>Dry Bulb and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 49.2  | 48.6  | 49.9  | 52.8  | 56.7  | 62.1  | 64.4  | 63.1  | 61.1  | 56.4  | 53.5  | 50.3 |
|                                                                               |             | MCWB   | 44.6  | 43.5  | 44.7  | 46.2  | 48.1  | 53.1  | 54.3  | 55.5  | 54.3  | 50.6  | 48.4  | 44.9 |
|                                                                               | 2%          | DB     | 46.5  | 46.2  | 46.2  | 48.6  | 52.6  | 58.6  | 61.1  | 59.8  | 58.3  | 52.5  | 48.5  | 46.9 |
|                                                                               |             | MCWB   | 42.3  | 42.0  | 42.0  | 43.4  | 46.3  | 50.9  | 53.4  | 53.3  | 52.6  | 47.0  | 43.9  | 42.4 |
|                                                                               | 5%          | DB     | 43.6  | 43.4  | 43.4  | 45.8  | 50.3  | 56.0  | 58.6  | 57.6  | 56.0  | 49.5  | 44.9  | 43.9 |
|                                                                               |             | MCWB   | 39.8  | 39.5  | 39.8  | 41.2  | 44.9  | 49.4  | 52.1  | 52.1  | 50.9  | 45.4  | 41.3  | 40.3 |
|                                                                               | 10%         | DB     | 40.9  | 40.5  | 40.4  | 43.4  | 48.1  | 54.0  | 56.6  | 55.6  | 53.6  | 47.0  | 42.4  | 41.3 |
|                                                                               |             | MCWB   | 37.4  | 37.1  | 37.0  | 39.6  | 43.6  | 48.3  | 51.0  | 51.1  | 49.1  | 43.8  | 39.5  | 38.3 |
| Monthly Design<br>Wet Bulb and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 45.3  | 44.4  | 45.0  | 46.8  | 49.7  | 54.0  | 56.6  | 56.8  | 55.4  | 51.1  | 48.6  | 45.8 |
|                                                                               |             | MCDB   | 48.8  | 47.1  | 49.1  | 52.0  | 55.2  | 59.8  | 62.1  | 61.2  | 60.1  | 56.5  | 53.1  | 50.3 |
|                                                                               | 2%          | WB     | 42.4  | 42.1  | 42.5  | 43.7  | 46.9  | 52.2  | 54.6  | 54.6  | 53.0  | 47.6  | 44.1  | 42.8 |
|                                                                               |             | MCDB   | 45.9  | 45.8  | 46.0  | 48.1  | 51.4  | 57.2  | 59.4  | 58.4  | 57.5  | 51.5  | 47.9  | 46.6 |
|                                                                               | 5%          | WB     | 40.0  | 39.7  | 40.0  | 41.6  | 45.4  | 50.4  | 53.0  | 53.0  | 51.3  | 45.8  | 41.8  | 40.6 |
|                                                                               |             | MCDB   | 43.3  | 43.1  | 43.2  | 45.4  | 49.6  | 55.0  | 57.3  | 56.5  | 55.4  | 49.1  | 44.6  | 43.6 |
|                                                                               | 10%         | WB     | 37.8  | 37.3  | 37.3  | 39.8  | 43.8  | 48.8  | 51.7  | 51.6  | 49.6  | 43.9  | 39.7  | 38.4 |
|                                                                               |             | MCDB   | 40.8  | 40.2  | 40.0  | 43.1  | 47.7  | 53.2  | 55.8  | 54.9  | 53.1  | 46.8  | 42.2  | 41.1 |
| Mean Daily<br>Temperature<br>Range                                            | 5% DB       | MDBR   | 7.2   | 7.6   | 7.6   | 7.7   | 8.1   | 8.7   | 8.6   | 8.1   | 7.4   | 6.4   | 7.0   | 7.5  |
|                                                                               |             | MCDBR  | 9.7   | 11.0  | 10.2  | 10.8  | 11.6  | 12.5  | 12.8  | 11.1  | 9.7   | 9.3   | 10.7  | 10.2 |
|                                                                               |             | MCWBR  | 8.1   | 9.0   | 8.3   | 7.6   | 7.7   | 7.6   | 7.4   | 7.1   | 7.2   | 7.2   | 9.0   | 8.3  |
|                                                                               | 5% WB       | MCDBR  | 9.6   | 10.8  | 10.0  | 10.7  | 10.6  | 11.0  | 11.1  | 9.9   | 9.3   | 8.7   | 10.2  | 10.1 |
|                                                                               |             | MCWBR  | 8.3   | 9.1   | 8.5   | 7.8   | 7.4   | 7.3   | 6.9   | 6.8   | 7.2   | 7.1   | 8.8   | 8.4  |
| Clear-Sky<br>Solar<br>Irradiance                                              | taub        | 0.388  | 0.291 | 0.319 | 0.346 | 0.347 | 0.337 | 0.347 | 0.344 | 0.337 | 0.288 | 0.366 | 0.380 |      |
|                                                                               | taud        | 2.532  | 2.143 | 2.260 | 2.272 | 2.394 | 2.476 | 2.476 | 2.502 | 2.416 | 2.255 | 2.648 | 2.632 |      |
|                                                                               | Ebn at Noon | 61     | 174   | 234   | 258   | 271   | 277   | 270   | 259   | 230   | 176   | 70    | 17    |      |
|                                                                               | Edh at Noon | 7      | 17    | 25    | 32    | 31    | 30    | 29    | 26    | 22    | 15    | 6     | 3     |      |
| All-Sky Solar<br>Radiation                                                    | RadAvg      | 19     | 114   | 399   | 879   | 1335  | 1609  | 1389  | 1049  | 561   | 188   | 35    | 3     |      |
|                                                                               | RadStd      | 1      | 19    | 47    | 81    | 155   | 196   | 135   | 103   | 47    | 20    | 3     | 0     |      |